

Final Comparison Table

Model	Val_Acc	Val_Prec	Val_Rec	Val_F1	Val_AUC	Test_Acc	Test_Prec	Test_Rec	Test_F1	Test_AUC
Logistic Regression	0.819060	0.818267	0.657573	0.729172	0.900350	0.819755	0.815955	0.662948	0.731537	0.898010
Decision Tree	0.837930	0.825541	0.713188	0.765262	0.917732	0.839019	0.825664	0.716762	0.767369	0.918256
Random Forest	0.788912	0.961812	0.447928	0.611208	0.919285	0.792730	0.965880	0.456587	0.620061	0.921883
Gradient Boosting	0.849654	0.839243	0.734891	0.783608	0.925033	0.849629	0.838051	0.736358	0.783920	0.926574
KNN	0.827713	0.780553	0.744084	0.761883	0.893025	0.834497	0.786450	0.759421	0.772699	0.897800
SVC3	0.820623	0.859605	0.616428	0.717985	0.890984	0.820705	0.861457	0.614863	0.717565	0.888524
Voting Ensemble	0.847309	0.866404	0.694951	0.771264	0.928204	0.845943	0.864672	0.692493	0.769063	0.929942
Bayesian Stacking	0.849375	0.818476	0.762472	0.789482	0.928490	0.849461	0.818608	0.762587	0.789605	0.930053